SASWAT DASH

Mechanical and Manufacturing Engineer

Kolkata, India | +91 88954 06527 | dashsaswat09@gmail.com | linkedin.com/in/saswatdash2003 | github.com/saswat-projects-hub

SUMMARY

Mechanical and manufacturing engineer from IIT Kharagpur (dual degree, CGPA 7.11, GATE 2024 qualified). M.Tech research raised the formability of magnesium sheet 55% through warm incremental forming; designed motion-control hardware at Podhikal Scientific; campus winner of Tata Elxsi TELIPORT. Now a Procurement Officer at HPCL, writing technical specifications and evaluating bids.

EXPERIENCE

Hindustan Petroleum Corporation Limited (HPCL) | *Procurement Officer* | Kolkata — Aug 2026 – Present

- Review indents from user departments and turn them into clear specifications, quantities and cost estimates
- Run techno-commercial evaluation of bids with engineering and finance teams and recommend awards
- Issue purchase orders and contracts, then track delivery, inspection and vendor payments to closure
- Manage vendor relationships and performance, keeping every step transparent and audit-ready

Podhikal Scientific Pvt. Ltd. | *Product Development Intern* | Scientific instruments — May 2023 – Jul 2023

- Modelled and optimised the XY motion-controller assembly in SolidWorks to minimise vibration
- Prototyped the motion-control unit in lightweight alloys, cutting weight 12%
- Designed a micro-fluid cap geometry to control droplet size and spray angle for uniform thin-film coating

Vecmocon Technologies | *Data Analyst & Strategy Intern* | EV intelligence — May 2024 – Jul 2024

- Analysed 50,000+ EV trip records in SQL to map demand, income and supply gaps for an NBFC leasing programme
- Correlated SONAR occupancy signals with BMS energy use to estimate passenger load and guide route allocation
- Improved rule-based route categorisation 27% using OpenStreetMap road tags, speed trends and stop density

Nexorea | *Co-Founder & Product Head* | AI mental-wellness startup — May 2025 – Present

- Stress-tested a mobile-first architecture for 6,000+ DAUs; prototypes hit under 200 ms latency, 35% faster than baseline
- Ran 48 user interviews and 7 design sprints over a 12-week pre-MVP to define an AI companion and diary

InstaPrepsAI | *Product Management & Analyst Intern* | B2B and B2C EdTech — Jun 2025 – Sep 2025

- Led a 6-member team through research, design and agile sprints to launch a Student Admin Assessment platform in 9 days
- Grew monthly active users 57% to 1.8k+ with a relaunch strategy built on behavioural analysis and A/B tests

Devlaunch | *Product Management Intern* | B2B SaaS — Apr 2025 – Jun 2025

- Cut bounce rate 12% by running 37 user interviews and turning the insights into PRDs that restructured navigation

RESEARCH AND PROJECTS

Warm forming of magnesium alloys, M.Tech research, IIT Kharagpur — Aug 2025 – 2026

- Developed warm Single Point Incremental Forming of magnesium sheet, raising formability 55% over room temperature
- Formed walls down to 0.4 mm thickness, with 25% finer grains under warm forming

Tata Elxsi TELIPORT, campus winner, 7,000+ teams nationally — Sep 2023 – Oct 2023

- Analysed GPS multipath interference across flight phases and proposed a tool suite to detect and filter it

American Express Campus Challenge, Decision Science track — Jun 2024 – Jul 2024

- Ranked 9th of 4,518 predicting T20 winners with XGBoost and CatBoost; SMOTE and tuning lifted accuracy 26%

AirtCraft, B2B art marketplace | Co-founder — 2024 – 2026

- Surveyed 481 artists and buyers, pivoted to B2B, and launched a quote-based platform with 4 role-based dashboards

EDUCATION

Indian Institute of Technology Kharagpur — 2021 – 2026

B.Tech (Hons.) Mechanical Engineering and M.Tech Manufacturing Science & Engineering, 5-year Dual Degree
CGPA 7.11/10 | Class XII, CHSE Odisha: 79.83% (2020) | Class X, BSE Odisha: 85.5% (2018)

SKILLS

Engineering: SolidWorks, AutoCAD, Fusion 360, Siemens NX, MATLAB, Mastercam, CNC, metal forming, failure analysis

Analytics: SQL, Python (Pandas, NumPy, SciPy, scikit-learn), Power BI, Tableau, Excel

Procurement and operations: Indent review, specifications, tendering on e-procurement portals, techno-commercial bid evaluation, purchase orders and contracts, vendor management, cost estimation, inventory analysis

ACHIEVEMENTS AND LEADERSHIP

- Campus winner, Tata Elxsi TELIPORT, a national competition of 7,000+ teams (2023)
- Qualified GATE 2024 (Mechanical Engineering)
- Rank 9 of 4,518, American Express Campus Challenge, Decision Science track (2024)
- Governor, Public Policy & Governance Society, IIT Kharagpur: led 86 members; reached 10,000+ people in 15+ cities
- LiGHT Head, Gopali Youth Welfare Society: led 13 centres serving 450+ students; raised ₹3L+ in the annual fundraiser
- Secretary, Culinary Arts Subcommittee, Technology Students' Gymkhana: ran logistics and budgets for 40+ events